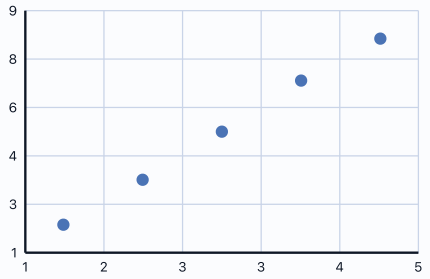
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Linear Regression & OLS Practice Exercises

1 MCQ

A fitted simple linear regression model is $\hat{y} = 7x + 18$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?



A 46

B 49

C 42

D 28

2 MCQ

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 20$ and the Total Sum of Squares is $\text{SST} = 100$. What is the coefficient of determination R^2?

A $R^2 = 0.80$

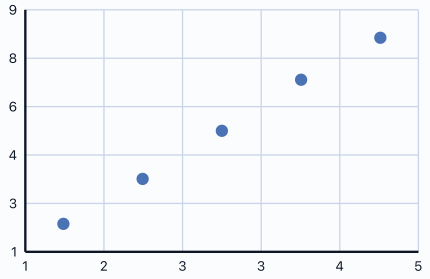
B $R^2 = 0.20$

C $R^2 = 0.90$

D $R^2 = 0.25$

3 MCQ

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 40$ and the Total Sum of Squares is $\text{SST} = 200$. What is the coefficient of determination R^2?



A $R^2 = 0.80$

B $R^2 = 0.20$

C $R^2 = 0.90$

D $R^2 = 0.25$

4 MCQ

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 60$ and the Total Sum of Squares is $\text{SST} = 300$. What is the coefficient of determination R^2?

A $R^2 = 0.80$

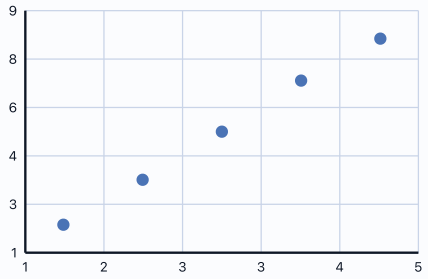
B $R^2 = 0.20$

C $R^2 = 0.90$

D $R^2 = 0.25$

5 MCQ

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 80$ and the Total Sum of Squares is $\text{SST} = 400$. What is the coefficient of determination R^2?



A $R^2 = 0.80$

B $R^2 = 0.20$

C $R^2 = 0.90$

D $R^2 = 0.25$

6 MCQ

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 100$ and the Total Sum of Squares is $\text{SST} = 500$. What is the coefficient of determination R^2?

A $R^2 = 0.80$

B $R^2 = 0.20$

C $R^2 = 0.90$

D $R^2 = 0.25$

7 MCQ

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 120$ and the Total Sum of Squares is $\text{SST} = 600$. What is the coefficient of determination R^2?

A $R^2 = 0.80$

B $R^2 = 0.20$

C $R^2 = 0.90$

D $R^2 = 0.25$

8 MCQ

For a bivariate feature dataset, $\text{Cov}(x, y) = 12$, $\text{Var}(x) = 4$, $\bar{x} = 5$, and $\bar{y} = 25$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

A $\hat{\beta}_1 = 0.3333333333333333,\ \hat{\beta}_0 = 10$

B $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 10$

C $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 25$

D $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 5$

9 MCQ

Consider a dataset of n observations (x_i, y_i) for $i = 1, \ldots, n$, where x_i are scalar predictor variables and y_i are scalar response variables. The ordinary least squares (OLS) estimates for the intercept β_0 and the slope β_1 of the linear model $y = \beta_0 + \beta_1 x + \epsilon$ are given by $\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Suppose we have a dataset where $n = 5$, $\sum x_i = 15$, $\sum y_i = 30$, $\sum x_i^2 = 55$, $\sum y_i^2 = 210$, and $\sum x_i y_i = 105$. A new data point (x_6, y_6) is added to the dataset. If the new regression line that minimizes the sum of squared errors for the $n + 1 = 6$ points is parallel to the original regression line (i.e., $\hat{\beta}_1^{\text{new}} = \hat{\beta}_1^{\text{old}}$), what is the value of y_6 in terms of x_6?

A $y_6 = 1.5x_6 + 0.5$

B $y_6 = 1.5x_6 + 1.5$

C $y_6 = 2x_6 + 0$

D $y_6 = 2x_6 + 1.5$

10 MCQ

Considering a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 40$ and the Total Sum of Squares is $\text{SST} = 200$. What is the coefficient of determination R^2?

A $R^2 = 0.80$

B $R^2 = 0.25$

C $R^2 = 0.20$

D $R^2 = 0.90$

11 MCQ

Examining a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 60$ and the Total Sum of Squares is $\text{SST} = 300$. What is the coefficient of determination R^2?

A $R^2 = 0.20$

B $R^2 = 0.90$

C $R^2 = 0.80$

D $R^2 = 0.25$

12 MCQ

For the configuration in a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 80$ and the Total Sum of Squares is $\text{SST} = 400$. What is the coefficient of determination R^2?

A $R^2 = 0.20$

B $R^2 = 0.80$

C $R^2 = 0.90$

D $R^2 = 0.25$

13 MCQ

Analyzing a multiple linear regression model, the Sum of Squared Errors is $\mathrm{SSE} = 100$ and the Total Sum of Squares is $\mathrm{SST} = 500$. What is the coefficient of determination R^2?

A $R^2 = 0.80$

B $R^2 = 0.90$

C $R^2 = 0.25$

D $R^2 = 0.20$

14 MCQ

Within a multiple linear regression model, the Sum of Squared Errors is $\mathrm{SSE} = 120$ and the Total Sum of Squares is $\mathrm{SST} = 600$. What is the coefficient of determination R^2?

A $R^2 = 0.80$

B $R^2 = 0.20$

C $R^2 = 0.90$

D $R^2 = 0.25$

15 MCQ

For a bivariate feature dataset, $\mathrm{Cov}(x, y) = 24$, $\mathrm{Var}(x) = 8$, $\bar{x} = 5$, and $\bar{y} = 30$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

A $\hat{\beta}_1 = \frac{1}{3},\ \hat{\beta}_0 = 15$

B $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 30$

C $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 15$

D $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 10$

16 MCQ

For a bivariate feature dataset, $\mathrm{Cov}(x, y) = 36$, $\mathrm{Var}(x) = 12$, $\bar{x} = 5$, and $\bar{y} = 35$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

A $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 35$

B $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 15$

C $\hat{\beta}_1 = \frac{1}{3},\ \hat{\beta}_0 = 20$

D $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 20$

17 MCQ

For a bivariate feature dataset, $\mathrm{Cov}(x, y) = 48$, $\mathrm{Var}(x) = 16$, $\bar{x} = 5$, and $\bar{y} = 40$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

A $\hat{\beta}_1 = \frac{1}{3},\ \hat{\beta}_0 = 25$

B $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 40$

C $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 25$

D $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 20$

18 MCQ

For a bivariate feature dataset, $\mathrm{Cov}(x, y) = 60$, $\mathrm{Var}(x) = 20$, $\bar{x} = 5$, and $\bar{y} = 45$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

A $\hat{\beta}_1 = \frac{1}{3},\ \hat{\beta}_0 = 30$

B $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 45$

C $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 30$

D $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 25$

19 MCQ

For a bivariate feature dataset, $\mathrm{Cov}(x, y) = 72$, $\mathrm{Var}(x) = 24$, $\bar{x} = 5$, and $\bar{y} = 50$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

A $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 50$

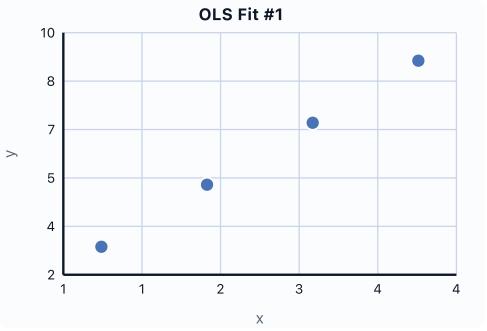
B $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 30$

C $\hat{\beta}_1 = \frac{1}{3},\ \hat{\beta}_0 = 35$

D $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 35$

20 MCQ

An Ordinary Least Squares regression model (#1) achieves residual sum of squares $\mathrm{SSE} = 5$ on data with total sum of squares $\mathrm{SST} = 20$. Compute the coefficient of determination R^2.



A $\frac{1}{2}$

B $\frac{1}{4}$

C $\frac{3}{4}$

D 3

21 MCQ

In dataset #2, sample variance is $s_x^2 = 3$, sample covariance is $s_{xy} = -8$, $\bar{x} = 3$, and $\bar{y} = 10$. Find the OLS slope $\hat{\beta}_1$.

A 14

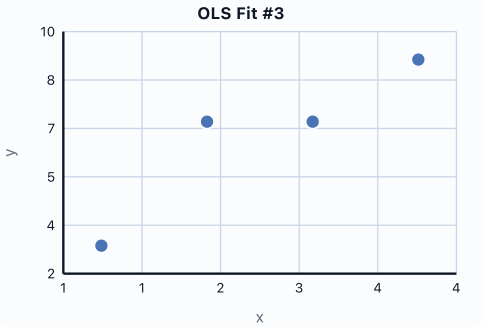
B $-\frac{3}{8}$

C $-\frac{8}{3}$

D $\frac{10}{3}$

22 MCQ

An Ordinary Least Squares regression model (#3) achieves residual sum of squares $\mathrm{SSE} = 15$ on data with total sum of squares $\mathrm{SST} = 70$. Compute the coefficient of determination R^2.



A $\frac{4}{7}$

B $\frac{11}{3}$

C $\frac{11}{14}$

D $\frac{3}{14}$

23 MCQ

In dataset #4, sample variance is $s_x^2 = 5$, sample covariance is $s_{xy} = -5$, $\bar{x} = 5$, and $\bar{y} = 10$. Find the OLS slope $\hat{\beta}_1$.

A -1

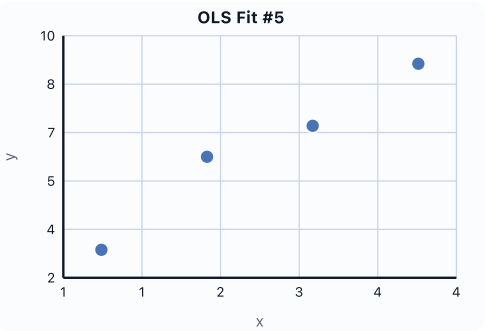
B 13

C 25

D 2

24 MCQ

An Ordinary Least Squares regression model (#5) achieves residual sum of squares $\mathrm{SSE} = 25$ on data with total sum of squares $\mathrm{SST} = 60$. Compute the coefficient of determination R^2.



A $\frac{1}{6}$

B $\frac{7}{12}$

C $\frac{7}{5}$

D $\frac{5}{12}$

25 MCQ

In dataset #6, sample variance is $s_x^2 = 7$, sample covariance is $s_{xy} = 9$, $\bar{x} = 2$, and $\bar{y} = 11$. Find the OLS slope $\hat{\beta}_1$.

A $\frac{9}{7}$

B $\frac{9}{2}$

C $\frac{7}{9}$

D $\frac{11}{2}$